

SOUGANDHI MANONMANI PENDYALA

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Summary

Software Development Engineer at Amazon with experience building scalable full-stack applications, distributed systems, and cloud-native services using Java, Spring Boot, React, TypeScript, and AWS. Skilled in designing event-driven architectures, REST APIs, automated testing frameworks, and operational monitoring solutions. Strong foundation in software engineering, distributed systems, data structures, and algorithms with experience delivering production-ready applications at scale.

Professional Experience

Software Development Engineer | Amazon, Detroit, MI, USA Sep '25 – Present

- Designed and delivered a full-stack seller migration feature using Spring Boot REST APIs and React, enabling merchants to manage BIL migration workflows with merchant access validation and achieving 100% unit test coverage.
- Implemented event-driven asynchronous offer expansion processing using AWS SQS FIFO queues, including queue consumers, filter-chain logic, dead-letter queue handling, and multi-tier CloudWatch alerting.
- Built a dynamic traffic-shaping system using AWS AppConfig and policy-based controls, enabling real-time rate-limit adjustments across AWS regions without service redeployments.
- Configured deployment pipelines and optimized Java microservices through JVM tuning, startup automation scripts, service health monitoring, and production-readiness enhancements.
- Developed a React-based cooldown mechanism with localStorage persistence and polling hooks, ensuring seamless state recovery and improved user experience across browser sessions.
- Authored unit tests using JUnit and Mockito, integration tests, and frontend functional tests using Vitest, achieving 98%+ code coverage across backend and frontend services.
- Participated in on-call rotations, resolved production incidents, performed root-cause analyses, and improved observability through diagnostic logging and monitoring enhancements.
- Leveraged AI-assisted development tools including Claude and Kiro to accelerate feature development, debugging, testing, and engineering productivity.

Education

Illinois Institute of Technology, Chicago, USA CGPA: 4.0/4.0
Master of Information Technology and Management

Kamala Institute of Technology and Science, Telangana CGPA: 3.0/4.0
Bachelor of Technology in Electronics and Communication Engineering

Skills

- Languages:** Java, TypeScript, JavaScript, Python, SQL, HTML, CSS
- Full Stack:** React, Redux, TypeScript, Spring Boot, REST APIs, Microservices, Distributed Systems
- Cloud & AWS:** AWS CDK, SQS, DynamoDB, CloudWatch, IAM, AppConfig
- Devops & Tools:** Docker, Kubernetes, CI/CD, Git, GitHub, GitLab, Jira, Claude, Kiro
- Testing:** JUnit, Mockito, Vitest, Integration Testing, Functional Testing
- Soft Skills:** Problem Solving, Communication, Cross-Functional Collaboration, Agile Methodologies
- Core CS:** Data Structures, Algorithms, Object-Oriented Programming

Publications

Improving Software Quality Framework Using AI Platforms Dec 2024 – Feb 2025

- Published a peer-reviewed book chapter analyzing the integration of AI tools such as Jira and CASE tools within Software Quality Assurance processes.
- Explored AI-driven defect detection, risk prediction, workflow automation, and Agile/DevOps support.

Projects

Multi Language AI Chatbot | Python, Streamlit :

Engineered the chatbot, using OpenAI's API, Google Translate API to provide real-time conversational capabilities. Implemented 5 language compatibility, secure user authentication with registration and login functionalities, with a chat history feature to store and retrieve user interactions.

Pawcare | Java, Android Studio :

Developed a full-service pet care application supporting grooming, veterinary appointments, and pet-sitting services through a unified platform. Implemented appointment scheduling, reminders, and service management workflows. Integrated Google Maps, Firebase, and Calendar APIs

Online Doctor Appointment Booking System | Java, JavaFX, MySQL :

Developed a JavaFX-based doctor appointment booking system using MVC architecture. Implemented authentication, profile management, appointment scheduling, and CRUD functionality. Integrated MySQL for persistent data management and application workflows.

E-Learning Platform for Students | Figma :

Designed an e-learning platform supporting both remote and on-campus students. Created six user-focused screens for communication, coursework tracking, and learning management workflows.

Certifications

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| • Codecademy Java Certification | • Google UX Design Specialization |
| • FreeCodeCamp Responsive Web Design | • Front-End Engineer (Codecademy) |
| • Jira Fundamentals Badge | • IBM Fundamentals of Project Management |